

explained in the following steps. If you are on an RPM-managed system, you can download and install the RPMs (see *Resources* section at the end of this article) rather than compile from source code.

```
$ wget http://download.openvz.org/utls/vzctl/3.0.23/src/vzctl-3.0.23.tar.bz2
$ tar xjf vzctl-3.0.23.tar.bz2
$ cd vzctl-3.0.23
$ ./configure && make && sudo make install

$ wget http://download.openvz.org/utls/vzquota/3.0.12/src/vzquota-3.0.12.tar.bz2
$ tar xjf vzquota-3.0.12.tar.bz2
$ cd vzquota-3.0.12
$ make && sudo make install
```

Now that both the kernel and the utilities are installed, it's time to reboot into the new kernel. Once you reboot the system, select the "OpenVZ" option from the GRUB menu and we are booting into the system with the shiny new OpenVZ kernel. We are now ready to create Virtual Environments (VE). It is possible to create VEs based on popular distributions.

We will take the simplest approach to create VEs, which is to use a template cache. This is nothing but a file containing an archive of a Linux root filesystem. This root file system may be based on Debian, Gentoo, Ubuntu, Fedora or any distro that you prefer. There are many template caches available from the OpenVZ website. Please see the *Resources* section at the end of the article. These template caches need to be copied into the location `/vz/template/cache` directory. Once they are, we can go about creating VEs based on these caches. Some template caches are available from the OpenVZ website, so let us download a few:

```
$ wget http://download.openvz.org/template/precreated/contrib/centos-5-1386-minimal-5.3-20090330.tar.gz
$ wget http://download.openvz.org/template/precreated/contrib/debian-5-0-1386-minimal.tar.gz
$ wget http://download.openvz.org/template/precreated/contrib/fedora-10-1386-default-20090318.tar.gz
$ wget http://download.openvz.org/template/precreated/contrib/ubuntu-7.10-1386-minimal.tar.gz
```

Now, let us create a VE based on Ubuntu:

```
# vzctl create 101 --os-template ubuntu-1.10-1386-minimal --config vps.basic
```

The `vzctl` command can be used for many different things in OpenVZ. In the above command line we are using it to create a VE with an ID "101". The `--os-template` option tells `vzctl` which template cache to use. It is basically the name of the template cache tar

file without the trailing `.tar.gz`. The `--config` option tells `vzctl` to use a base config file called `vps.basic`. This file contains VE parameters that specify various limits and barriers among other things. There are several values in the `vps.basic` file that are applied to the VE after creating it. But these values can be overridden by specifying them explicitly for this VE. For example, we will look at how to set various values for the VE we just created, using the same `vzctl` command:

```
# vzctl set 101 --onboot --save
```

This command sets the "onboot" parameter of VE 101 to `true`. This means that the VE will be automatically started when the hardware node boots up. Each VPS has a private configuration file under `/etc/vz/conf/<VE-ID>.conf`. When you specify the `--save` option to `vzctl`, the setting is applied to the VE, if it is running and is also saved to the VE's private configuration file.

```
# vzctl set 101 --hostname ubuntuve.k7computing.com --save
# vzctl set 101 --ipadd 192.168.1.101 --save
# vzctl set 101 --nameserver 192.168.1.1 --save
```

These commands set and save VPS networking parameters. It is now time to start the VPS:

```
# vzctl start 101
```

Let's view the list of VEs by running the `vzlist` command:

```
# vzlist
VPSID  NPROC STATUS IP_ADDR  HOSTNAME
101    20 running 192.168.1.101 ubuntuve.k7computing.com
```

Setting the root password and logging in via SSH

Since the VPS has an IP address, you can use SSH to log into it. But the VE must be running the SSH daemon and you must know the root password. There are multiple ways to set the root password and we shall look at two of them:

- Setting the password using the `vzctl` command

```
# vzctl set 101 --userpasswd root.test123
```

- Entering into a VE

```
# vzctl enter 101
ubuntuve# passwd
```

Changing password for user root.

New UNIX password:

Retype new UNIX password:

passwd: all authentication tokens updated successfully.

The `vzctl enter <ve-id>` command is used to "enter" into a VPS. You can then run commands as if you have logged into the VE itself. Of course, this is only possible